

Hari Priya P B

Software Engineer

itsharipriya8919@gmail.com | [LinkedIn](#) | [Portfolio](#) | Manchester, United Kingdom

SUMMARY

Experienced and driven Software Engineer with 4+ years of professional experience building production systems across healthcare, e-pharmacy and enterprise domains. Specialised in Ruby on Rails and Java, with hands-on expertise in microservices, RESTful APIs, event-driven architecture, workflow automation and system integration on AWS, and the ability to adapt across diverse technical and business contexts. Skilled in mentoring junior engineers, fostering cross-team collaboration and delivering business-critical applications, while supporting systems in production. Enjoys tackling complex, real-world problems, finding creative ways to simplify them, and building technology that makes a meaningful difference to people. Further strengthened by an MSc in Advanced Computer Science at the University of Manchester, with a focus on Machine Learning and AI and hands-on projects in NLP, LLMs, RAG and computer vision.

TECHNICAL SKILLS

- **Languages:** Python, Ruby, Java, SQL, JavaScript
- **Frameworks:** Ruby on Rails, Spring/Spring Boot, Flask
- **Databases & Persistence:** PostgreSQL, MySQL, MongoDB, Redis, Database Design, Query Optimisation, ORM, JPA, Hibernate, JDBC, Active Record
- **Architecture & Design:** Microservices, Event-Driven Architecture, Distributed Systems, RESTful APIs (JSON), MVC, OOP/OOD, SOLID, Design Patterns
- **Messaging & Async:** Apache Kafka, SNS, SQS, Sidekiq, Karafka
- **Cloud, DevOps & Observability:** AWS (EC2, S3, RDS, DMS, MSK, CloudWatch), Docker, Kubernetes, Jenkins, CI/CD, Linux, Bash, Sentry, New Relic, Kibana
- **Testing:** JUnit, Mockito, RSpec, TDD, Unit Testing, Integration Testing
- **Tools & Collaboration:** Git, Bitbucket, Postman, Swagger/OpenAPI, Jira, Confluence, Agile/Scrum
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, NLP, Computer Vision, Generative AI, LLMs, VLMs, RAG, Embeddings, Vector Databases (FAISS, pgvector), Fine-tuning (PEFT/LoRA), Model Evaluation

PROFESSIONAL EXPERIENCE

Software Development Engineer

Oct 2022 - Jun 2025

Tata 1mg - India's Largest E-pharmacy & Digital Health Platform

Bangalore, India

- Built and maintained backend services for ODIN, Tata 1mg's in-house warehouse management platform, using **Ruby on Rails**, **PostgreSQL** and **AWS** in a **microservices** setup, supporting inventory, orders and purchasing across multiple warehouses and retail stores.
- Cut response times of high-traffic APIs by 75% by optimising **ActiveRecord** queries and removing unnecessary database calls.
- Improved inventory reconciliation accuracy by 25%+ across 50+ vendors by processing each vendor's updates in order using **AWS SQS FIFO** queues, with **Redis** tracking failed updates for retry.
- Automated purchase-order processing by connecting ODIN to the finance system (FAS) and SAP through **RESTful APIs**, handling 500+ B2B orders a month, and used **Sidekiq** and **Apache Kafka** to update expired orders and keep other services in sync.
- Built an audit-logging pipeline using **AWS DMS**, **MSK** and **Karafka** that records every database change (what changed, who changed it and when) across multiple tables in one central history.

- Automated the Re-GRN (goods return) workflow and sped it up with bulk **SQL** inserts, reducing manual effort by 40%+ and saving 15+ minutes per return.
- Built backend automation for a direct Picker-to-Packer handover, saving Rs.6 lakhs/month (~£4.6K), and contributed to the company-wide SKU barcoding rollout, which saved Rs.20 lakhs (~£15.5K) in manpower costs.
- Worked in an **Agile/Scrum** team using **Jira** and **Git/Bitbucket**, writing **RSpec** tests, resolving 15+ production incidents a month with **Sentry**, **New Relic** and **Kibana**, and mentoring 3+ junior engineers.

Software Engineer

Jul 2021 - Oct 2022

Tata Digital Health - Telemedicine Platform

Bangalore, India

- Built backend **RESTful APIs** using **Java**, **Spring Boot** and **Spring Data JPA/Hibernate** with **MySQL**, within a **microservices** architecture for Tata Digital Health's telemedicine platform, supporting patient consultations and digital prescription (Rx) issuance and retrieval for 1,000+ consultations per day.
- Developed Clinical Decision Support System (CDSS) features, including demographic-based symptom prompts and language-based doctor matching across 4+ languages, reducing consultation setup time by 15%.
- Integrated **Single Sign-On (SSO)** and appointment-reminder push notifications, and contributed to the paid e-consultation cart-to-order and payment flow, supporting the patient journey from login through booking and payment.

Software Engineering Intern

Feb 2021 - Jun 2021

Cognizant - Global Technology Services & Consulting Company

India

- Developed a full-stack application end to end, building the backend in **Java** and **Spring Boot** with a layered **MVC** design and integrating it with the frontend, applying **OOP**, **SOLID** and **TDD** with **JUnit** and **Mockito** to deliver modular, well-tested features.
- Led a 4-member team through **Agile/Scrum** sprints, owning task planning, stand-ups and delivery, and was selected for a return offer as Programmer Analyst Trainee based on performance.

PROJECTS

NutriVision - AI Dietary Assessment

- Fine-tuned **BLIP-2 (Flan-T5-XL)**, a multimodal vision-language model (**VLM**), for automated dietary assessment from food images using **LoRA** with **PyTorch**, **Hugging Face Transformers** and **PEFT**.
- Benchmarked three LoRA fine-tuning configurations across five tasks on the Nutrition5K dataset, running a structured **model evaluation** to compare accuracy and efficiency trade-offs.
- Designed a zero-cost inference-time post-processing pipeline that improved dietary reasoning accuracy without additional training or parameters, and built a **Gradio** demo to showcase results.

Healthcare Monitoring Using Machine Learning - Published, IJPRSE

- Built a full-stack web application using **Python**, **Flask** and **scikit-learn** to predict patient health-risk levels from biometric inputs (age, BMI, blood pressure, pulse rate, temperature).
- Trained a Decision Tree classifier with **Pandas** and **NumPy** for data cleaning and preprocessing, serialised the model with **joblib**, and served real-time predictions through a **Bootstrap 4** web interface.

EDUCATION

MSc Advanced Computer Science

Sep 2025 - Sep 2026

University of Manchester

Manchester, United Kingdom

Global Future Scholarship Recipient

BTech in Information Technology

Aug 2017 - Apr 2021

Sona College of Technology

Tamil Nadu, India

CGPA: 8.92 / 10